

AI Agents That Actually Work

A 20-minute field guide for developers

Why this talk now

- Agent demos look magical on stage
- Production agents quietly fall over
- The gap is architecture, not prompts
- 20 minutes to close it



IMAGE:stage-vs-production

Why most agents fail

- Prompt engineering hits a ceiling
- No persistent memory
- Tool use is brittle
- No recovery path when steps go wrong



IMAGE:agents-failing

The three pillars of a working agent

Planning, memory, and tool use — each load-bearing, each a common failure point.



SMARTART:three-pillars

Pillar 1 — Planning

- Decompose goals into verifiable steps
- Budget tokens and tool calls per step
- Always leave a retry path
- Let the plan be inspected, not implicit



IMAGE:planning-tree

Pillar 2 — Memory

- Short-term: scratchpad for the current task
- Long-term: retrievable facts + episodes
- Evict aggressively; relevance beats recall
- Write-through, never write-only



IMAGE:memory-layers

Pillar 3 — Tool use

- Narrow, typed, well-named tools
- One-shot success rate > tool count
- Return structured errors, never stack traces
- Treat tools as contracts, not endpoints



SMARTART:tool-contract

Case study: the invoice agent

- Demo: flawless. Production: 38% failure.
- Root cause: no memory of prior invoices
- Fix: episodic memory + typed tools
- Result: 38% -> 4% after one sprint



IMAGE:case-study-chart

What to do on Monday

- Instrument your agent's plan — make it visible
- Add one memory layer, measure recall accuracy
- Rewrite your three worst tool schemas
- Pick one failure mode per sprint, kill it



IMAGE:monday-checklist

Ship agents that actually work

Plan visibly. Remember deliberately. Use tools like contracts.

Questions? @your-handle